Xgpu

GPU Extension for RISC-V

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Chapter 1. Operation

1.1. BAR.SYNC

Synchronise a group of warps inside a block

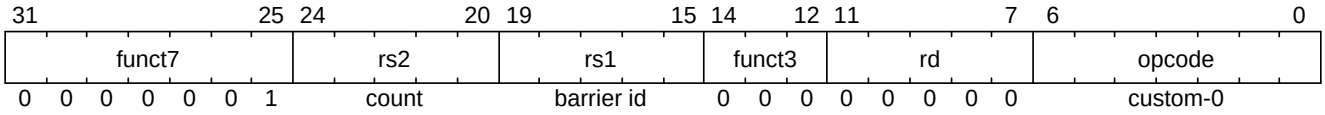


Figure 1. Encoding of BAR.SYNC

Table 1. Fields of BAR.SYNC

Field	Bits	Value	Meaning
opcode	6:0	0001011	custom-0, opcode of all xgpu-commands. Specified by RISC-V.
rd	11:7	00000	Reserved, has to be x0. Different values trigger an illegal-instruction-exception.
funct3	14:12	000	Chooses BAR.SYNC in the barrier commands.
rs1	19:15	*reg*	Register with barrier id to enable multiple barriers at once.
rs2	24:20	*reg*	Register with count of how many warps are required.
funct7	31:25	0000001	Barrier group in custom-0.

1.1.1. Functionality

A warp that executes BAR.SYNC increases the counter of the barrier X[rs1] by one and is held, until the counter reaches the value in X[rs2]. All warps are then released and the counter is set back to 0 for supporting loops. rs1 and rs2 have to be the same for all threads, else it's undefined bahaviour

1.1.2. Edge Cases

The counter can't be higher than the number of wards and is higher than 0. Else it's undefined behaviour.

A warp that is held by two barriers is released, only if both barriers release it.

Store: A warp is raising the count by 1, only if the store data + address is in the store buffer

Divergence: If the warp is still in diverged state and reaches the barrier, that's undefined behaviour

1.1.3. Example

Double Buffering: In the initializing phase, all warps load data in the shared memory, not all with the same pace. Warp A loads the first half, warp B the second one. If warp A finishes faster, it would start computing with data, warp B has yet not written.
With barriers the warps wait for all data to be written.